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Review work on nanoparticle carriers for drug delivery -- How the physicochemical properties of certain proteins, polymers, and ligands can influence the loading and release profile of a certain drug, exploring protein-protein and protein-ligand interactions.

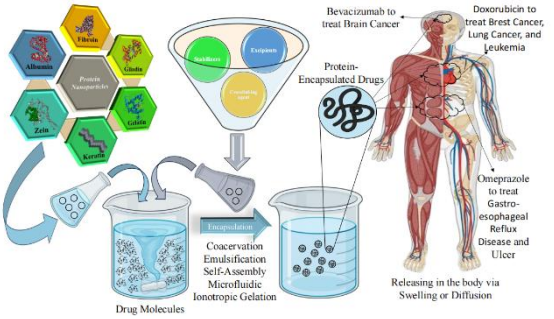
Two review articles written a year apart, regarding drug delivery systems. One works from the inside out: which proteins make a usable carrier, and how the encapsulation route decides what the payload does once it is in. The other works from the outside in: how a chitosan surface is modified so the particle is recognized by the cells it was meant for.

Insights: Physicochemical Properties of Protein, Encapsulation methodologies, Loading Parameters, Release Kinetics, Drug delivery.

Encapsulation of Protein-Based Drug Carriers for Controlled Loading and Releasing Efficiency: A Review.

The Protein Journal (Springer Nature), accepted June 2026. doi:10.1007/s10930-026-10346-4

Mahadesh Chandro Mondal, H.M. Wasi Uddin, *et al.*
My contribution: Conceptualization, Analysis & Formatting, Visualization, Pictorial, Writing-Review & Editing.

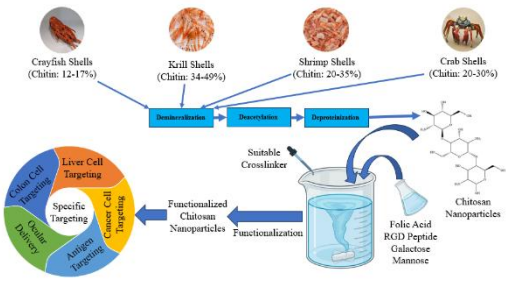


Among the eight protein carriers reviewed, the encapsulation efficiency ranges between 42-96%, while loading capacity ranges much wider, from ~2% to 75%. In this study, gliadin, Type-B gelatin, and albumin were at the upper end (reported values of approximately 80–98%), and BSA and Gliadin showed higher loading capacity (reported values exceeding 20%). However, in my observation, high encapsulation efficiency does not necessarily mean effective drug delivery. For example, a Paclitaxel-loaded gelatin-methacryloyl system by Vigata *et al.* showed about 96% encapsulation efficiency, yet 25% of the drug remained trapped after 3 months of cumulative drug release. Release tracks structure more than efficiency. In a study, Ferritin's 12–30 nm cages leak little at pH 7.4, but release 55.6-72.5% of cargo within 8 h once acidified, and silk fibroin can be tuned from 5% to 90% release purely through β-sheet crystallinity. Among methodologies, coacervation and emulsification are used widely for their simplicity, though only emulsification is both highly scalable and highly reproducible. However, microfluidics provides the highest possible size control and seems to me the strongest prospect for future drug and gene delivery.

Insights: Physicochemical Properties and Synthesis of Chitosan Nanoparticles, Surface Functionalization, Ligand Conjugation, Ligand-Polymer Interaction, Targeted Drug Delivery.

Functionalization and Ligand Conjugation of Novel Chitosan Nanoparticles for Targeted Drug Delivery Systems: A Review.

Manuscript Under Review
Mahadesh Chandro Mondal, Arpita Dutta, *et al.*
My contribution: Conceptualization, Analysis & Formatting, Visualization, Pictorial, Writing-Review & Editing.



Across the four ligands reviewed, encapsulation efficiency for conjugated chitosan systems sits at 60–90%, but the more telling variable is surface charge: mannose conjugation drops zeta potential from +32.2 to +25.4 mV, while RGD-functionalized micelles hold +38.7 ± 1.2 mV at 267.3 nm with 92.5% encapsulation and a 7.2-fold increase in doxorubicin uptake. In my observation, targeting gain rather than loading is what distinguishes these systems; hyaluronic acid-modified nanoparticles carrying 5-fluorouracil proved nearly 60-fold more effective against HT-29 cells than the free drug, at unremarkable encapsulation efficiency. Comparing conjugation methods, ionic gelation gives the highest efficiency at the lowest cost (100–400 nm, 70–90% EE, no organic solvent), emulsion crosslinking trades efficiency for mechanical stability (150–600 nm, 60–85%), and spray drying scales best but imposes high thermal stress (200–500 nm, 60–80%). Microfluidics yields the smallest and most uniform particles (20–300 nm) and seems to me the most promising route, though scalability remains its limiting constraint.

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Full text and the draft manuscript available on request